with a general explanation on how to make each circuit work. We'll leave the specifics up to you for further exploration.

MOSFETs

The Metal Oxide Field Effect Transistor, or MOSFET, is one of the simpler components to use with DC motors along with the basic transistor shown in Figure 6-4 of Chapter 6. The following circuit in Figures 11-4 and 11-5 uses an IRL540 MOSFET, although others would work as well, to switch a motor with a load of up to about 20 amps. That's a pretty big motor! The IRL540 can also switch at a very fast rate of speed, so we can use PWM to control the speed of a motor connected in this circuit. This circuit also uses a 1N4001 diode connected to the positive and negative side of the motor to prevent any electrical surges from the motor damaging the MOSFET.

■ **Note** In all of these motor circuits, we have a pin that is marked +V MOTOR. While we have connected these to our standard +5 VDC supply because our motors worked on +5 volts, you may need to connect these pins to a larger power supply, depending on the needs of your motor.

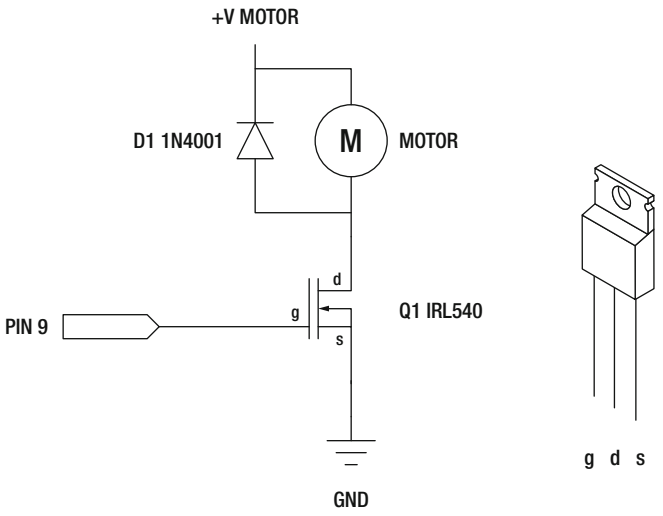

Figure 11-4. *MOSFET and DC Motor schematic*